Reg. No. | | | | | | | | | | | [14EC62]

SIXTH SEMESTER B.Tech EXAMINATION, JUNE – 2019

INFORMATION THEORY AND CODING

Time: 3 Hours **Maximum Marks: 70**

Instructions:

Answer any FIVE full questions by choosing at least TWO questions from each part

PART-A

1. **a.** Show that the entropy of the nth extension of a discrete memory less source is n times the entropy of the original source. **05**

b. A black and white TV picture consists of 525 lines of picture information. Assume that each line consists of 525 picture elements and each element can have 256 brightness levels. If pictures are repeated at the rate of 30 frames/sec, calculate the average rate of information conveyed by the TV set to a viewer. **05**

c. State the properties of mutual information. Represent the relation between mutual information and entropy by Venn diagram. **04**

2. **a.** Consider a Discrete Memory less Source X with symbols x_i, i=1,2,3,4. Table below lists four possible binary codes. **07**

x_i	Code A	Code B	Code C	Code D
x_1	00	0	0	0
x_2	01	10	11	100
x_3	10	11	100	110
x_4	11	110	110	111

i. Show that all the codes except code B satisfy the Kraft inequality.

ii. Show that codes A and D are uniquely decodable but codes B and C are not uniquely decodable.

b. A DMS X has five symbols x_1, x_2, x_3, x_4 and x_5 with $P(x_1)=0.4$, $P(x_2) = 0.19$, $P(x_3)=0.16$, $P(x_4)=0.15$ and $P(x_5)=0.1$

i. Construct a Shannon-Fano code for X, and calculate the efficiency of the code

ii. Repeat for the Huffman code and compare the results. **07**

3. **a.** Define the following terms and their significance **04**

i. Entropy

ii. Self-information

iii. Information rate

iv. Source efficiency

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b. For the channel matrix given below, compute the channel capacity

$$P(Y/X) = \begin{bmatrix} 0.6 & 0.2 & 0.2 \\ 0.2 & 0.6 & 0.2 \\ 0.2 & 0.2 & 0.6 \end{bmatrix} \text{ with } r_s = 1000 \text{ symbols/sec.}$$ **05**

A binary channel has the following characteristics

$$P(Y/X) = \begin{bmatrix} 2/3 & 1/3 \\ 1/3 & 2/3 \end{bmatrix}$$

c. If input symbols are transmitted with probabilities ¾ and ¼ respectively, find H(X), H(X,Y) and H(Y/X). **05**

4. a. Consider a continuous random variable having a distribution as given below **04**

$$f_x(x) = \frac{1}{a}, \quad 0 \le x \le a$$

$$= 0, \text{ otherwise}$$

Find the differential entropy H(x).

b. State the capacity theorem for continuous channels.

A voice grade channel of a PSTN has a bandwidth of 3.4 kHz.

Calculate

i. capacity of the channel for a SNR = 30 dB

ii. Minimum SNR to support information transmission through the channel at a rate of 4800 bits/sec. **06**

c. A Gaussian channel has a 10 MHz bandwidth and a two sided noise power spectral density($\eta/2$) of $10^{-14} W/$ Hz. The signal power at the receiver has to be maintained at a level less than or equal to 1/100 of a milliwatt. What is the capacity of this channel? **04**

PART- B

5. a. Discuss the error controlling techniques for communication channels. **04**

b. Define the terms Linear and Cyclic with respect to error correcting codes. **02**

c. Consider a linear code with generator matrix $\begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$.

Find

i. all code vectors **08**

ii. (ii)all the Hamming weights and distances

iii. (iii)minimum weight parity check matrix

iv. (iv)Draw the encoder circuit for the above codes.

6. a. Give two properties and advantages of cyclic codes. **04**

b. Find the product of $(1 + x + x^3)$and $(1 + x + x^2 + x^4)$ using modulo-2 algebra. **04**

c. Construct a triple error correcting BCH code with a block length of 31. **06**

7. a. Consider a (3,1,2) convolution code with $g^{(1)} = (1\ 1\ 1)$, $g^{(2)} = (1\ 0\ 1)$ and $g^{(3)} = (1\ 1\ 1)$

i. Draw the encoder block diagram

ii. Find the generator matrix

iii. Find the code word corresponding to the information sequence (1 1 1 0 1) using time domain and transform domain approach. **10**

b. Discuss Trellis diagram with an example. **04**

8. a. Distinguish between: Random errors and Burst errors.

If a (15, 9) cyclic code generated by $g(x) = 1 + x^3 + x^4 + x^5 + x^6$ has a burst error correcting ability of 3, what will be its burst error correcting efficiency. **04**

b. Construct a (15,13) single error correcting RS code and determine the systematic code word corresponding to the information polynomial $i = (0\ 0\ \propto\ 0\ 0\ 1\ \propto^7\ \propto^2\ \ 0\ 0\ 1\ \propto\ \propto^2)$ where $\propto$ is a primitive element of $GF(2^4)$. **10**